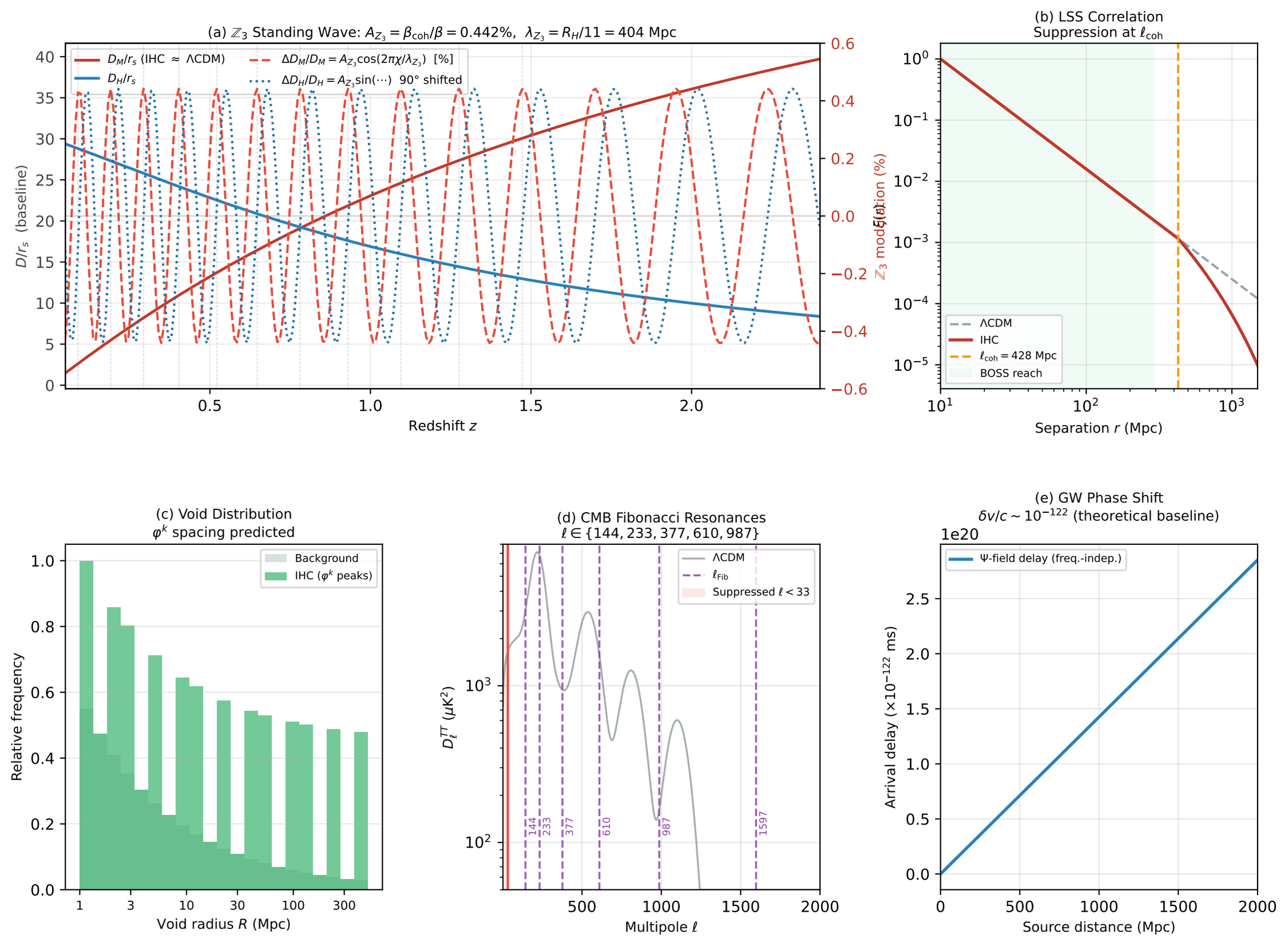
(a) $\mathbb{Z}_3$ Standing Wave: $A_{Z_3} = \beta_{coh}/\beta = 0.442\%$, $\lambda_{Z_3} = R_H/11 = 404$ Mpc
D_M/r_s (IHC ≈ ΛCDM)
D_H/r_s
$\Delta D_M/D_M = A_{Z_3}\cos(2\pi\chi/\lambda_{Z_3})$ [%]
$\Delta D_H/D_H = A_{Z_3}\sin(\cdots)$ 90° shifted
D/r_s (baseline)
Redshift z
(b) LSS Correlation
Suppression at ℓ_{coh}
$\mathbb{Z}_3$ modulation (%)
ΛCDM
IHC
$\ell_{coh} = 428$ Mpc
BOSS reach
Separation r (Mpc)
(c) Void Distribution
φ^k spacing predicted
Background
IHC (φ^k peaks)
Relative frequency
Void radius R (Mpc)
(d) CMB Fibonacci Resonances
$\ell \in \{144, 233, 377, 610, 987\}$
ΛCDM
ℓ_{Fib}
Suppressed $\ell < 33$
D_ℓ^{TT} (μK^2)
144
233
377
610
987
1597
Multipole ℓ
(e) GW Phase Shift
$\delta v/c \sim 10^{-122}$ (theoretical baseline)
1e20
Ψ-field delay (freq.-indep.)
Arrival delay (×10^{-122} ms)
Source distance (Mpc)